Discrete Random Variables & the Binomial Distribution

ANSWER KEY

Probability distributions and expected value

1 A game costs 2 dollars to play. You win 10 dollars with probability 0.1, and win nothing otherwise. Let X be your net profit. Find the probability distribution of X and its expected value.

$X = 8$ (win 10, minus the 2 cost) with probability 0.1; $X = -2$ (lose the 2 cost) with probability 0.9. Expected value: $E(X) = 8(0.1) + (-2)(0.9) = 0.8 - 1.8 = -1$. On average, a player loses 1 dollar per game.

2 Find the standard deviation of the profit X from the previous problem, to two decimal places.

Variance $= \sum (x-\mu)^2 P(x) = (8-(-1))^2(0.1) + (-2-(-1))^2(0.9) = (81)(0.1) + (1)(0.9) = 8.1 + 0.9 = 9$. Standard deviation $= \sqrt{9} = 3$.

Binomial setting and conditions

3 State the four conditions required for a random variable to follow a binomial distribution (sometimes remembered as BINS: Binary, Independent, Number fixed, Same probability).

(1) Each trial has only two possible outcomes (success/failure). (2) Trials are independent. (3) There is a fixed number of trials, n. (4) The probability of success, p, is the same for every trial.

4 A survey randomly selects 20 people (without replacement) from a small town of 50 people and records how many support a proposal. Explain why this technically violates the independence condition, and why it is often still treated as approximately binomial.

Sampling without replacement from a small population means each selection slightly changes the probabilities for the next draw, violating strict independence. However, when the sample size is small relative to the population (a common guideline is the sample being less than 10% of the population), the effect is negligible, and the situation is treated as approximately binomial — though here 20 out of 50 is 40%, so this would NOT satisfy the 10% condition and should be treated with caution.

Binomial probability formula

5 A basketball player makes 70% of free throws. Find the probability she makes exactly 4 out of 6 attempts, using $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$.

$P(X=4) = \binom{6}{4}(0.7)^4(0.3)^2 = 15(0.2401)(0.09) \approx 0.324$.

6 Using the same free-throw shooter, find the mean and standard deviation of the number of makes out of 6 attempts, using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.

$\mu = np = 6(0.7) = 4.2$ makes. $\sigma = \sqrt{np(1-p)} = \sqrt{6(0.7)(0.3)} = \sqrt{1.26} \approx 1.12$ makes.

Visualizing a binomial distribution

7 The graph shows a smoothed approximation of the binomial distribution for $n = 20$ trials with $p = 0.5$, plotted as a continuous curve for visual comparison to a normal shape.

a Using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, verify that this binomial distribution has mean 10 and standard deviation approximately 2.24, matching the curve shown. $\mu = np = 20(0.5) = 10$ ✓. $\sigma = \sqrt{np(1-p)} = \sqrt{20(0.5)(0.5)} = \sqrt{5} \approx 2.24$ ✓.

b Explain why a binomial distribution with $n = 20$ and $p = 0.5$ looks approximately bell-shaped, referencing the conditions for a normal approximation. A binomial distribution can be approximated by a normal distribution when both $np \geq 10$ and $n(1-p) \geq 10$. Here $np = 10$ and $n(1-p) = 10$, both meeting the threshold, so the shape is approximately bell-shaped/symmetric rather than skewed.

Geometric distribution (brief introduction)

8 A basketball player makes free throws with probability 0.7 each attempt, independently. Find the probability that her first miss occurs on the 3rd attempt (i.e. she makes the first 2, then misses the 3rd). $P(\text{first miss on attempt 3}) = (0.7)(0.7)(0.3) = 0.147$, using the geometric distribution's structure (successes then the first failure).

Applying the binomial distribution in context

9 A multiple-choice quiz has 10 questions, each with 4 choices, and a student guesses randomly on every question. Find the probability the student gets at least 8 questions correct, given that $P(X = 8) \approx 0.0004$, $P(X = 9) \approx 0.00003$, and $P(X = 10) \approx 0.000001$. $P(X \geq 8) = P(X = 8) + P(X = 9) + P(X = 10) \approx 0.0004 + 0.00003 + 0.000001 \approx 0.00043$, or about 0.043% — an extremely small chance, illustrating how unlikely it is to do well on a multiple-choice test by pure guessing.

Binomial probability of at least one success

10 A manufacturing process produces defective items independently with probability 0.05. In a batch of 20 items, find the probability that at least one item is defective, using the complement of "zero defective." $P(\text{zero defective}) = (0.95)^{20} \approx 0.358$. $P(\text{at least one defective}) = 1 - 0.358 = 0.642$.

11 Using the mean and standard deviation formulas $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, find the mean and standard deviation of the number of defective items in a batch of 20, with $p = 0.05$. $\mu = np = 20(0.05) = 1$. $\sigma = \sqrt{np(1-p)} = \sqrt{20(0.05)(0.95)} = \sqrt{0.95} \approx 0.975$.

Comparing binomial scenarios

12 Explain why a binomial model would NOT be appropriate for counting the number of aces dealt when drawing 5 cards without replacement from a standard 52-card deck, and name the distribution that would be more appropriate instead.

Drawing without replacement means each draw changes the composition of the remaining deck, so the probability of drawing an ace changes from draw to draw — violating the binomial requirement of a constant probability of success on every trial. This scenario is better modeled by the hypergeometric distribution, which accounts for sampling without replacement from a finite population.